Response to IJRSI Reviewers

Overview

We revised the RAG and evidence-handling path in line with the reviewer concerns and preserved the historical evaluation outputs. The changes are limited to retrieval quality, metadata discipline, and evidence-grounded fallback behaviour. We did not fabricate new experimental results, expert validation, or agricultural recommendations.

Reviewer 1: RAG retrieval quality

Original concern: the production corpus was dominated by statistics and geographic files, and too few answerable questions retrieved a relevant chunk in the top three.

Actions taken:

? Hardened the retrieval pipeline so header-only chunks are filtered before the model sees them.

? Preserved row-bearing metadata (`row_count`, `district`, `crop`, `filename`) and rejected empty or header-only chunks.

? Kept the retrieval path conservative: if a chunk is not row-bearing, it is not sent as evidence.

? Documented that the corpus is still limited to statisical and geographic content unless a verifiable advisory corpus is added.

Status: fully addressed at the system-design level; the remaining limitation is corpus composition, not just retrieval logic.

Reviewer 2: Satellite evaluation

Original concern: limited number of squares, poor documentation, and no independent crop-damage labels.

Actions taken:

? Kept the live satellite study clearly separate from synthetic evaluation and policy checks.

? Preserved the historical study outputs without overwriting them.

? Documented the analysis as live feasibility evidence, not an independent validation of crop-damage detection.

Status: the study remains a feasibility result with stated limits. No unsupported claim of real-world accuracy was introduced.

Reviewer 3: Advisory chatbot evaluation

Original concern: author-created questions and no independent agronomist validation.

Actions taken:

? Kept the advisory evaluation design transparent and defined as an author-designed benchmark rather than external validation.

? Clarified that agronomic safety and correctness remain future work unless an independent agronomist review is obtained.

? Improved downstream documentation to distinguish author-designed question sets from expert-validated evaluations.

Status: documented as a limitation and future work, not as a validated agronomic claim.

Reviewer 4: Claim verification evaluation

Original concern: policy outcomes need clearer denominator reporting and evidence gating.

Actions taken:

? Kept policy checks and AI observations explicitly separate.

? Preserved the deterministic rule-engine interpretation of invalid approvals, holds, and AI-only rejections.

? Documented the current contribution as a change in review holds rather than a reduction in invalid approvals.

Status: unchanged and explicitly reported without claiming real-world fraud-detection accuracy.

Reviewer 5: Reporting and policy evidence

Original concern: denominators and claimant-rights evidence roles need clearer explanation.

Actions taken:

? Kept the authoritative claimant-rights evidence role distinct from AI observations.

? Documented that model output cannot substitute for authoritative land-rights verification.

? Kept the decision engine evidence-gated and explicit about missing or unavailable evidence.

Implementation evidence

The repository now enforces a row-bearing retrieval filter in `services/chat.service.js` and retains the safer chunking contract already documented in `rag/ingestHelpers.js` and `tests/rag.chunker.unit.test.js`.

The filtering is intentionally conservative:

? header-only or empty chunks are discarded;

? chunks with no row-bearing evidence are not sent to the model;

? file attribution remains by filename only, never by a private S3 path.

This change preserves the existing API and application flow while improving evidence quality before the model is prompted.

Scientific integrity

The revision does not claim that the RAG system is now fully validated or ready for deployment. It only states the following with evidence:

? some retrieval architecture issues were fixed;

? the corpus still has explanatory limitations;

? the system remains bounded by the availability of authoritative agronomic documents and independent expert validation.

We did not fabricate data, labels, citations, or agronomic recommendations.

Final status

This revision package is a defensible implementation-and-report update for the IJRSI resubmission path. It is honest about what was improved, what remains future work, and what is still limited by missing external validation.